

EE10510: Fundamentals of Electrical & Electronics Engineering
Assignment Sheet-I
Based on Unit II (Three-Phase Circuits)

Max. Marks: 5
Due date: 31.10.2025

1. A motor has an output of 4.8kW, an efficiency of 88% and a power factor of 0.635 lagging when operated from a 240V, 50Hz supply. It is required to improve the power factor to 0.95 lagging by connecting a capacitor bank in parallel with the motor. Calculate (i) current taken by motor (ii) supply current after power factor correction (iii) the current drawn by capacitor (iv) the value of capacitor (v) the kvar rating of the capacitor
2. A 230V, 50Hz single phase supply feeds following loads (i) incandescent lamps of taking current of 10A at unity power factor (ii) fluorescent lamps of taking 8A at a power factor of 0.7 lagging (iii) a 3kVA motor operating at full load and power factor of 0.8 lagging (iv) a static capacitor. Determine for the lamps and motor (a) total current (b) overall power factor and (c) total over (d) find the value of the static capacitor to improve the overall power factor of 0.975 lagging.
3. A three phase 415V has a delta connected load with $Z_{AB} = 10 \angle 0^\circ \Omega$ $Z_{BC} = 10 \angle 30^\circ \Omega$ and $Z_{CA} = 15 \angle -30^\circ \Omega$. Calculate the (a) load phase current (b) load line current (c) Active and reactive power per phase (d) three phase apparent power.

